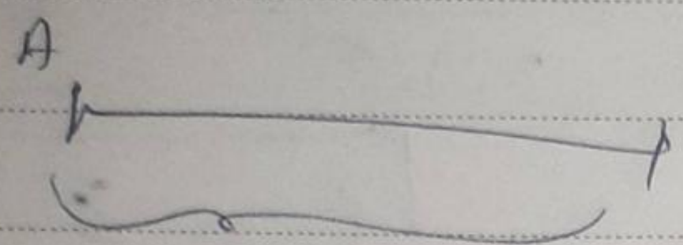
Subject:
Date

Problem 2

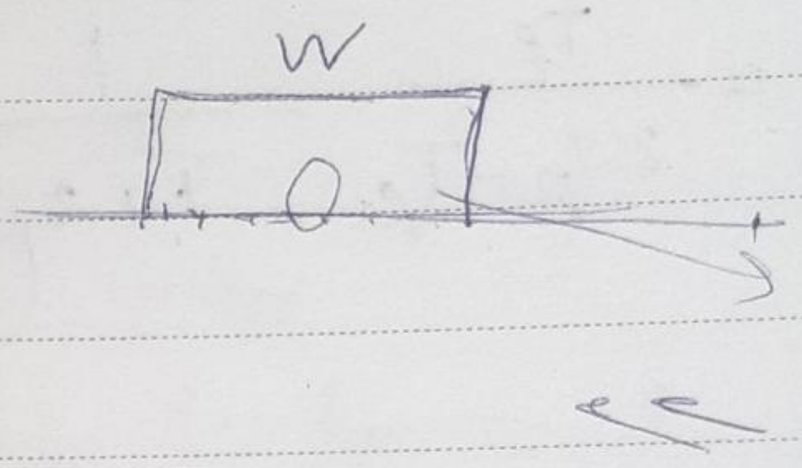
R bps

$V_{Signal} = V \frac{Km}{S}$



Frame = B bits

L_{km}



$$T_{propagation} = \frac{L}{V}$$

W

$$t_f = \frac{B}{R}$$

$$a = \frac{T_{propa}}{T_{frame}} = \frac{LR}{VB}$$

$$2a + 1 = \frac{2LR}{VB} + 1$$

$$N = \left\lceil \frac{2LR}{VB} + 1 \right\rceil$$

$$n = \lceil \log_2(N) \rceil$$

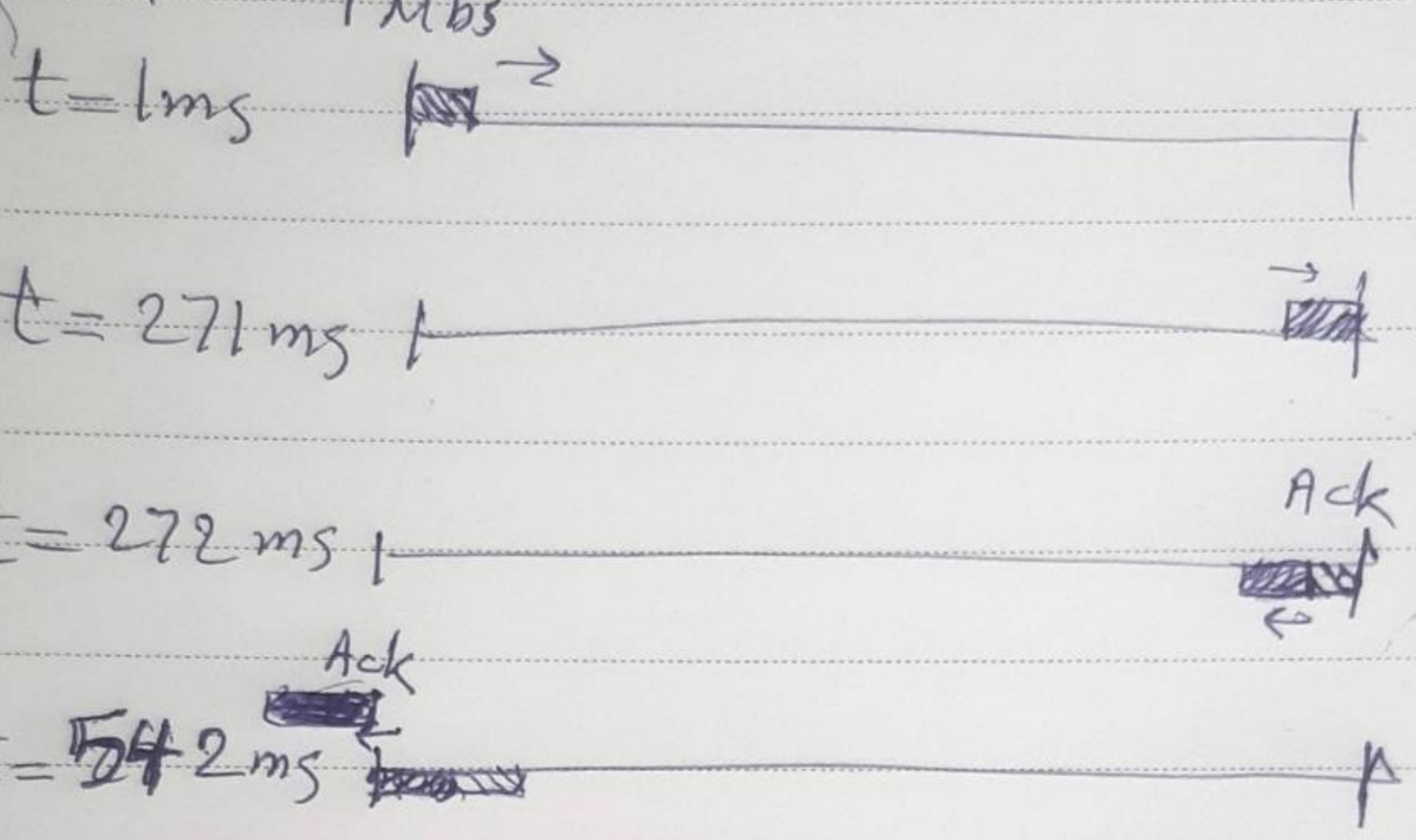
Subject:
Date

Problem 3

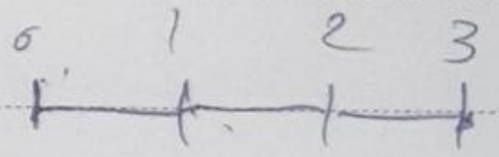
1000 bit $R = 1Mbps$

$T_p = 270ms$

$T_f = \frac{1000b}{1Mbs} = 1ms$



Stop & wait $U = \frac{1}{542ms} = 18\%$



* نکته: به Piggyback پردازی است.

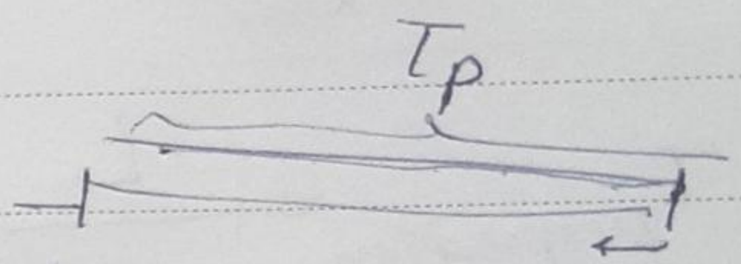
Subject:
Date

Problem 4

$$L = 9 \times 10^{10}\ m \quad \left.\begin{array}{l} \\ \end{array}\right\} \rightarrow \frac{L}{C} = \frac{9 \times 10^{10}}{3 \times 10^{8}} = 300\ S$$
$$C = 3 \times 10^{8}\ m/S$$

$$\boxed{t_p = 300\ S}$$

$$U_{S\&W} = \frac{T_f}{2T_p + T_f} = \frac{1}{1 + 2\frac{T_f}{T_p}}$$



a

نرخ بیت

$$\rightarrow \begin{cases} 300\ S \times \frac{64\ Mb}{S} = 19200\ Mb \quad \boxed{19.2\ Gb} \\ \frac{32 \times 8 \times k}{} \times \frac{S}{64 \times 10^{3}} = \boxed{4\ ms} \end{cases}$$

$$= \frac{1}{1 + 2\left(\frac{300}{4 \times 10^{-3}}\right)}$$

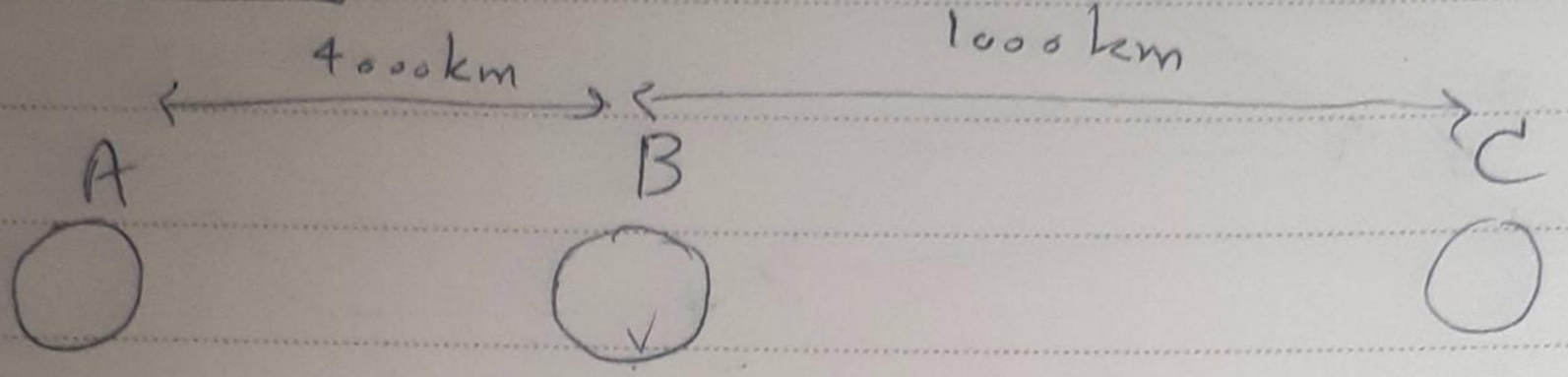
$$T_f = \frac{32 \times 8\ kb}{64000\ \frac{kb}{S}} = \boxed{4\ ms}$$

$$U = \frac{1}{1 + 2\frac{T_f}{T_p}} = \frac{1}{1 + 2\left(\frac{4\ ms}{300\ S}\right)} = \boxed{6.67 \times 10^{-4}}$$

Subject :

Date

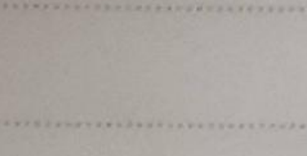
Problem 5



$R_{AB} = 100kbps$

$F = 1000b$

$W_{AB} = 3$



$R_{BC} = ?$

$T_{P_{AB}} = \frac{1000}{100kbps} = 10ms$

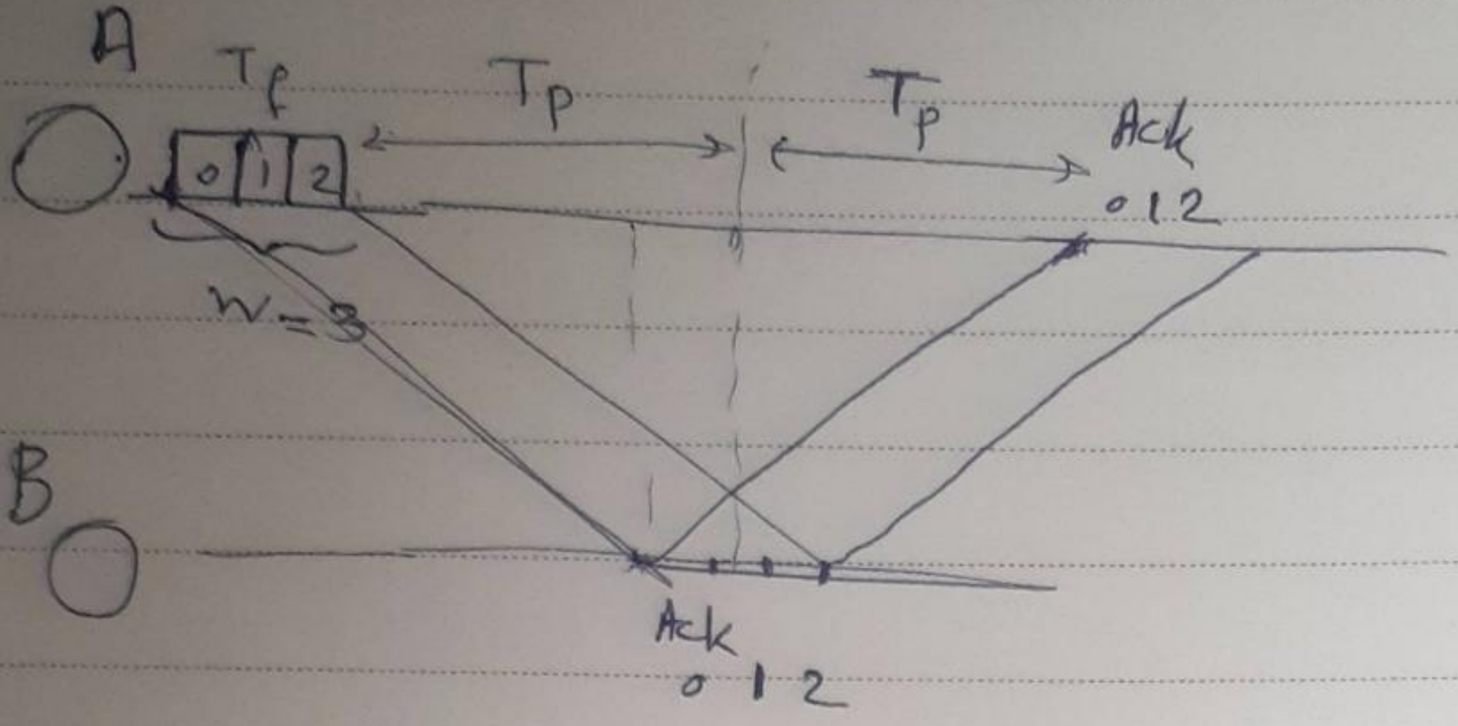
$T_{prop_{AB}} = \frac{4000km}{3 \times 10^8 m/s} = \frac{4}{300} s$

$T_{prop_{BC}} = \frac{1000km}{3 \times 10^8 m/s} = \frac{1}{300} s$

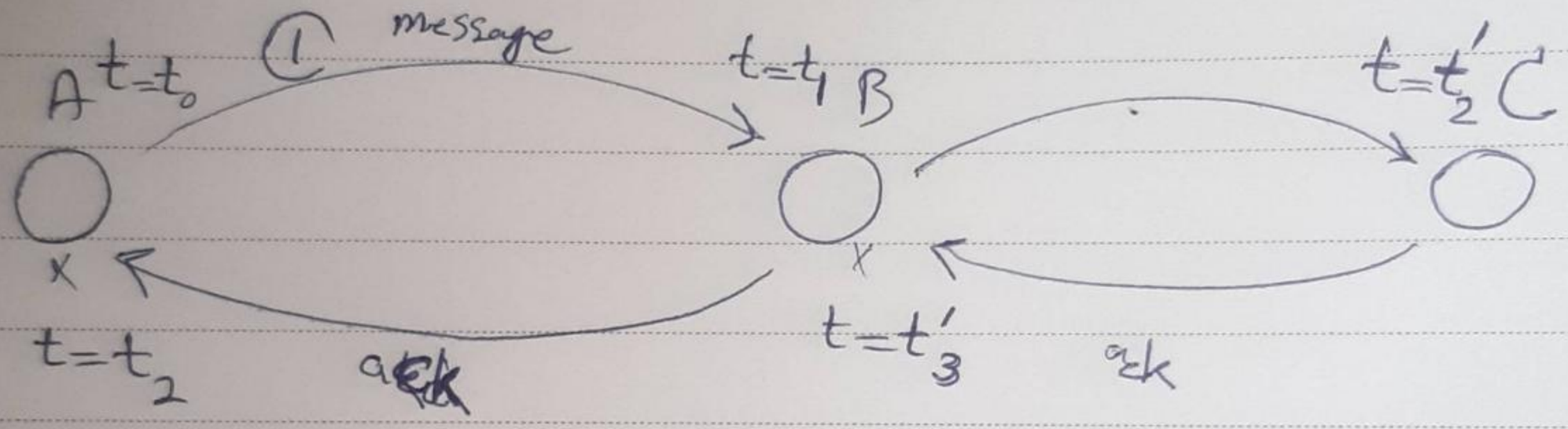
$T_{P_{BC}} = ? \quad \boxed{\frac{1000}{R}}$

Subject:
Date

Problem 5



الف) t



نکته ① مستقل بودن ack های پیام های دو تا کانکشن

$t = T_{final}$

② باید حداقل نرخ مورد نیاز را پیدا بکنیم.

$R \geq R_{min}$

$t_2 = t_3'$ باید

$$3\left[T_{f_{BC}} + 2T_{p_{BC}}\right] = T_{f_{AB}} + 2T_{p_{AB}}$$

$$3\left[\frac{1000}{R} + 2\left(\frac{1}{300}\right)\right] = 10ms + 2\left(\frac{4}{300}\right) \rightarrow \boxed{R = 180kbps}$$